Exploring Facial Feature Extraction and Temporal Dynamics for Deepfake Video Detection: A CNN-RNN Study on FaceForensics++

Umang Jeswani
Computer Science and Engineering
Jaypee University of Engineering and Technology
Guna, India
umangjeswani03@gmail.com

Dr. Priyadharshini S
Computer Science and Engineering
Indian Institute of Information Technology
Kottayam, India
priyadharshini@iiitkottayam.ac.in

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***Abstract*—The rise of generative artificial intelligence has led to the creation of deepfake videos—synthetically altered content where a person's appearance or voice is manipulated using deep learning. While these technologies have creative potential, their misuse poses serious risks including mis-information, identity theft, and privacy violations. Increasing access to deepfake tools makes it harder to distinguish real from manipulated content, undermining digital trust. This paper explores hybrid deep learning models that integrate spatial and temporal features for deepfake detection. CNNs are used to extract frame-wise facial features, while RNNs such as GRUs and LSTMs model temporal dynamics. Three architectures are implemented and compared: CNN + GRU, ResNeXt50 + LSTM, and XceptionNet + GRU. The models are trained on the FaceForensics++ dataset and tested on a curated set of ten videos. The ResNeXt50 + LSTM model achieved the best results, with 80% accuracy, 0.75 F1-score, and 0.80 ROC-AUC, demonstrating its ability to capture both appearance based and sequential patterns. This work highlights the importance of temporal modeling in deepfake detection and contributes toward building scalable, robust solutions for applications in multimedia forensics and digital authentication.**

***Index Terms*—Deepfake detection, Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), Temporal Modeling, FaceForensics++, Video forensics, GRU, LSTM**

I. Introduction

The deepfakes are generated using advanced machine learning techniques, such as Generative Adversarial Networks (GANs). These manipulations can realistically alter a person's facial expressions, voice, and identity, making fake videos nearly indistinguishable from authentic ones [1]. As the threat landscape grows, effective deepfake detection has become a critical research challenge. Traditional approaches primarily focused on frame-level analysis, using handcrafted features or shallow CNN-based architectures. Models such as MesoNet and XceptionNet aimed to detect subtle inconsistencies in individual frames [3], [4]. However, temporal patterns that emerge across video sequences, key indicators of manipulated content, are not explored [2], [5].

To address these limitations, recent work has explored hybrid spatiotemporal architectures combining Convolutional Neural Networks (CNNs) with Recurrent Neural Networks (RNNs) to process sequential patterns [2], [6], [7].

In this study, we evaluate three CNN–RNN-based models for deepfake video detection, namely CNN with GRU, ResNeXt50 with LSTM, and XceptionNet combined with GRU. These models are trained on the FaceForensics++ dataset [1] and tested on a curated collection of real and fake videos. The aim of this research is to identify a model that effectively captures spatiotemporal inconsistencies.

II. Related Works

Deepfake generation has rapidly progressed due to deep learning techniques like autoencoders and GANs. Early systems employed face-swapping, while newer tools such as StyleGAN and DeepFaceLab generate highly realistic faces using adversarial training [1], [2]. Software like Face2Face and FaceSwap enables real-time facial reenactment, and datasets like DeepFake-TIMIT have further enhanced audio-video realism, complicating detection [3]. Initial detection approaches relied on frame-level analysis with handcrafted features or shallow CNNs. Lightweight models like MesoNet targeted compression artifacts, whereas XceptionNet, adapted from image classification, detected subtle inconsistencies [4], [5]. However, these methods struggled with temporal artifacts such as flickering and unnatural expressions [6].

Recent work integrates spatial and temporal modeling using hybrid architectures. CNNs extract frame-level features, while RNNs like LSTMs and GRUs capture temporal dynamics. Guera and Delp proposed a CNN–LSTM pipeline for this purpose [7], followed by recurrent CNNs [8] and 3D CNNs with multimodal learning [9], offering improved robustness under compression and varying conditions.

Standardized datasets are vital for training and evaluation. FaceForensics++ (FF++) includes manipulations using Deepfakes, FaceSwap, and NeuralTextures across various quality levels [1]. This study uses FF++ for training and a curated Deepfake Detection Challenge (DFDC) subset for testing. Nonetheless, current models often lack generalization across manipulation types and datasets. Temporal anomalies like

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inconsistent lip-sync or blinking are frequently overlooked. Moreover, real-time detection on low-resource devices is underexplored [11].

To address these issues, this work evaluates three spatiotemporal deep learning models, namely, CNN with GRU, ResNeXt50 with LSTM, and XceptionNet combined with GRU. These models are trained and tested in a controlled setting to assess their effectiveness in improving deepfake detection accuracy and reliability.

III. METHODOLOGY

A. Dataset Used

The dataset used is FaceForensics++ (FF++) C23 compressed dataset, obtained from Kaggle [1]. It contains high-quality real and fake videos. To assess model generalization on unseen data, a custom test set was curated using samples from the DFDC dataset [6]. The test set consisted of ten short video clips, five real and five fake, manually selected and organized into separate folders for binary classification. These videos were not part of the FF++ training set, ensuring unbiased evaluation.

Each video, whether from the training or test set, underwent a standardized preprocessing pipeline. First, frames were extracted at a constant frame rate to preserve temporal consistency. The frames are of size 224 x 224, which were then grouped into fixed-length sequences of 20 consecutive frames, enabling the system to learn temporal inconsistencies, such as unnatural facial transitions or inconsistent blinking, common in deepfakes. This representation was crucial for supporting spatiotemporal learning in the detection models [2], [3].

B. Steps Involved in Deepfake Detection

The complete pipeline for deepfake detection is shown in Fig. 1, which outlines the core stages from data handling to final prediction. This modular workflow enables consistent processing across all model variants while leveraging both spatial and temporal features critical for accurate detection.

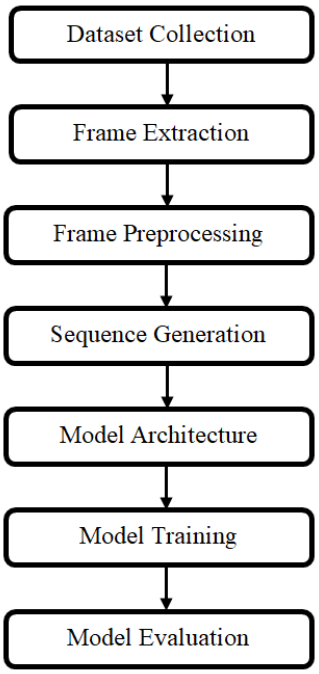


Fig. 1: Workflow.

The deepfake detection workflow implemented in this research comprises several systematically organized stages that collectively contribute to robust and reliable classification. The process begins with dataset collection, where the FaceForensics++ (FF++) C23 dataset was used as the primary training source [1]. This dataset includes real and manipulated videos generated using Face2Face, FaceSwap, Deepfakes, and NeuralTextures. Custom test data was curated from the Deepfake Detection Challenge (DFDC) dataset [6] to evaluate the trained model. This test set consisted of ten manually labelled short video clips, equally split between real and fake samples, and was excluded from the training and validation processes.

Following data collection, each video was subjected to frame extraction at a fixed frame rate, ensuring temporal consistency across the dataset. The extracted frames were preprocessed. This standardization helps improve model convergence and ensures uniform input dimensions. Subsequently, the pre-processed frames were segmented into 20 consecutive frames per sample. This helps to preserve motion continuity and enable the models to capture temporal inconsistencies, an essential factor in distinguishing real videos from deepfakes.

The core of the pipeline involved model design, where three distinct hybrid deep learning architectures were developed. The first architecture combined a shallow CNN for spatial feature extraction with a GRU (Gated Recurrent Unit) layer to model temporal dependencies. The second model utilized a pre-trained ResNeXt-50, followed by a Long Short-Term Memory (LSTM) [5], [12]. The third architecture paired the XceptionNet model [4] with GRU layers to integrate temporal modeling into the feature space.

Finally, in the model evaluation stage, predictions were generated at the sequence level for each test video. To determine the final video-level classification, a majority voting strategy was adopted, aggregating all sequence predictions. This ensemble-based technique proved effective in mitigating the influence of incorrectly predicted sequences, resulting in more stable and reliable classification outcomes [17], [18].

IV. EXPERIMENTAL RESULTS

Table I presents the predicted labels from three deepfake detection models, CNN with GRU, XceptionNet with GRU, and LSTM with ResNeXt50, evaluated on five representative video samples from a custom test set comprising 10 videos (5 real, 5 fake) extracted from the DFDC dataset [6]. Each model was assessed based on its ability to correctly classify each video as either real or fake. As observed in Table I, the LSTM with the ResNeXt50 model demonstrates the most reliable performance, correctly identifying 4 out of 5 videos, encompassing both real and fake classes.

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Notably, this model accurately classified all real videos and 2 out of 3 fake videos, indicating a balanced sensitivity and generalization capability across both classes. In contrast, the XceptionNet with the GRU model, although effective in detecting real videos, misclassified all three fake videos (Videos 3, 4, and 5) as real. This outcome suggests a significant bias toward the real class, limiting its utility for identifying manipulated content. The CNN with the GRU model performed the weakest among the three, with only 2 correct predictions. It misclassified two fake videos and erroneously labeled one real video as fake, reflecting lower overall accuracy and inconsistency in predictions.

These observations are consistent with the confusion matrices illustrated in Fig. 2, which further confirm that the LSTM with ResNeXt50 model achieves the most balanced and effective deepfake detection among the three architectures evaluated.

In Fig. 2(a), the CNN with GRU model correctly classified 4 out of 5 real videos and only 1 out of 5 fake videos, yielding a total of 5 correct and 5 incorrect predictions. The high misclassification of fake videos indicates poor discriminative power for detecting deepfakes in this model.

In contrast, Fig. 2(b) shows that XceptionNet with GRU correctly identified all 5 real videos but misclassified 4 out of 5 fake videos as real, achieving 6 correct and 4 incorrect predictions. While this model exhibits a stronger bias toward the real class, it slightly improves upon the CNN with GRU.

Fig. 2(c), representing the LSTM with ResNeXt50 model, demonstrates the most balanced performance, with 5 real and 3 fake videos correctly classified, resulting in 8 correct and 2 incorrect predictions.

The results outlined in Table II and visualized in Fig. 3 highlight the critical role of combining deep spatial representations with efficient temporal modeling in enhancing deepfake detection. Models equipped with shallow spatial backbones, such as CNN with GRU, failed to capture complex temporal artifacts and exhibited poor generalization. Meanwhile, even sophisticated spatial models like XceptionNet showed subpar performance when temporal modeling was weak or poorly balanced. In contrast, the ResNeXt50 with LSTM model effectively leveraged both spatial and temporal features.This suggests that future detection systems should prioritize architectures that integrate deep convolutional extractors with sequential learning mechanisms.

The key evaluation metrics used are summarized in Table II. The CNN with GRU model exhibited limited capacity for learning temporal dependencies, as indicated by its low recall of 0.20 and a ROC-AUC score of 0.24. These results suggest weak generalization and poor detection of fake samples, likely due to the simplicity of the spatial backbone and underdeveloped temporal representation. In contrast, the ResNeXt50 with LSTM model delivered the best overall performance. It achieved an accuracy of 0.80, an F1-score of 0.75, and a ROC-AUC of 0.80. These strong results, clearly reflected in both Table II and the trends observed in Fig. 3, demonstrate the effectiveness of combining deep spatial feature extraction with robust sequence modeling. Lastly, the XceptionNet with GRU model, while achieving perfect precision (1.00), suffered from low recall (0.20) and a moderate ROC-AUC of 0.32. As seen in Table II and illustrated in Fig. 3, this model lagged in overall balanced performance, indicating a tendency to overfit to real samples and an inability to effectively discriminate between real and fake classes.

Based on the comparative analysis, the ResNeXt50 with LSTM model emerged as the most effective among the evaluated architectures. It consistently outperformed CNN with GRU and XceptionNet with GRU across all evaluation metrics, as presented in Table II, and maintained a clear advantage in visual performance trends shown in Fig. 3.

V. Conclusion

This study implemented and evaluated three hybrid deep learning architectures namely, CNN with GRU, ResNeXt50 with LSTM, and XceptionNet with GRU, for deepfake video detection using the FaceForensics++ dataset, with evaluation performed on a curated test set of real and fake videos using multiple classification metrics. It is observed that, ResNeXt50 with LSTM outperformed the other models.The CNN with GRU model exhibited limited generalization and weak temporal learning, while XceptionNet with GRU achieved perfect precision but suffered from low recall, indicating a tendency to overfit. And ResNeXt50 with LSTM highlights the superior metric distribution compared to the other models, reinforcing that the temporal modeling significantly enhances deepfake detection when integrated with deep CNN backbones. However, certain limitations were noted in this study, including the use of a relatively small test set of only 10 videos, the exclusive focus on visual (frame-based) features without considering audio or other modalities.

Future work may address these limitations by testing on diverse datasets incorporating multimodal signals like audio-visual inputs, exploring Transformer-based or 3D CNN architectures for spatiotemporal learning, and optimizing models for real-time detection. Overall, this study demonstrates the value of hybrid spatiotemporal models in detecting manipulated video content in video forensics.

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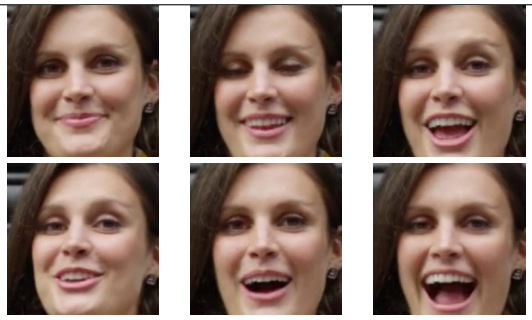
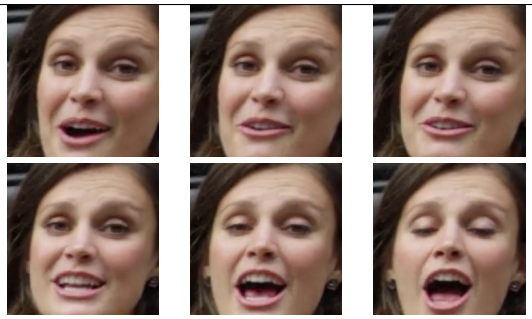
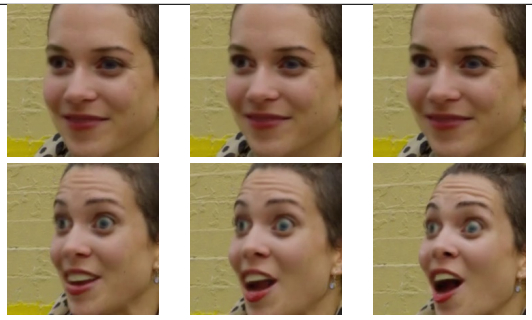
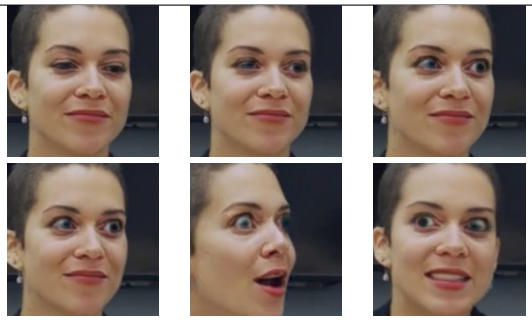
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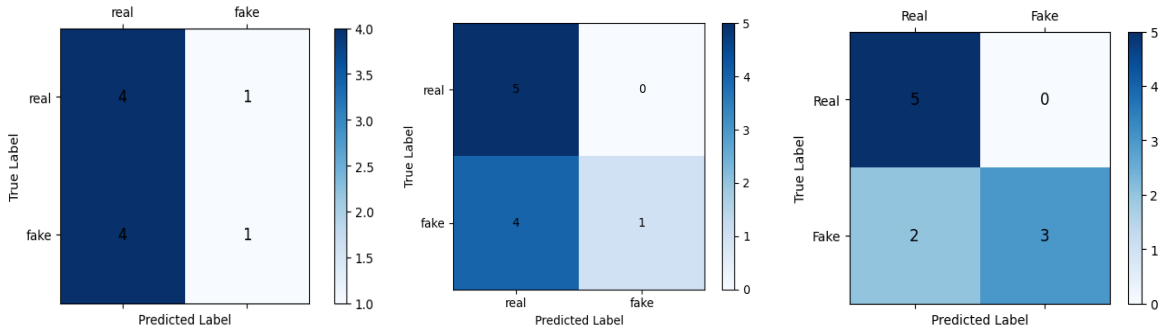
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TABLE I: Predicted labels

Input	Input Frames	Ground Truth	CNN with GRU	XceptionNet with GRU	LSTM with RestNext50
Video - 1		Real	Real	Real	Real
Video - 2		Real	Fake	Real	Real
Video - 3		Fake	Real	Real	Fake
Video - 4		Fake	Fake	Real	Fake
Video - 5		Fake	Real	Real	Fake

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(a) (b) (c)

Fig. 2: Confusion matrix (a) CNN with GRU (b)XceptionNet with GRU (c) LSTM with RestNext50

TABLE II: Evaluation metrics

Model	Metrics				
	Accuracy	***Precision***	***Recall***	***F1-Score***	***ROC-AUC***
CNN+GRU	0.5	0.5	0.2	0.29	0.24
ResNeXt50 with LSTM	0.8	1.0	0.6	0.75	0.8
XceptionNet with GRU	0.6	1.0	0.2	0.33	0.32

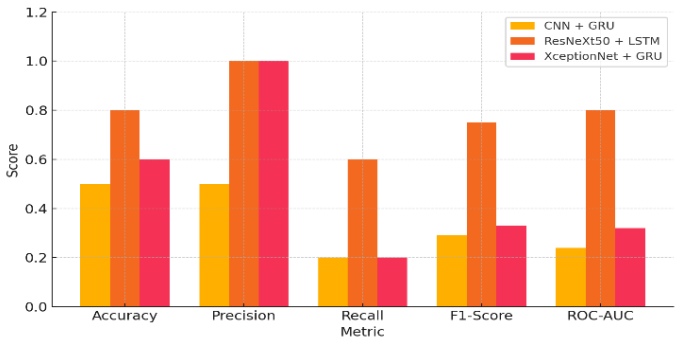


Fig. 3: Comparison of model performance across evaluation metrics.

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